SynthFlood: Ambient Synthetic Content Buys Web-Agent Decisions — A Dose–Response Analysis and a Provenance-Ordering Remedy

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CCS CONCEPTS

• **Information systems** → **World Wide Web**; • **Computing methodologies** → *Artificial intelligence.*

KEYWORDS

web agents, synthetic content, ambient contamination, dose–response, provenance, e-commerce

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1 INTRODUCTION

Web agents have crossed from demos to deployment: they shop, book, and compare on our behalf, turning the Web's product pages from artifacts that *inform humans* into signals that *direct machine decisions.* At the same time, the supply side of those pages is being rewritten. An estimated 27.5% of tokens passing common quality filters on the open Web are now AI-generated, and the share is climbing monthly [22]; measurement studies document machine-generated catalogs flooding music platforms [26] and scientific publishing [17]. This content is not adversarial in the classical sense: it is produced by ordinary commercial incentives — search-optimized content farms, counterfeit-style listing clones, review inflation — and a research program now studies how to *optimize* content for generative engines [1, 15, 27] and how to detect the result [6, 12, 34]. The consequence is that agentic systems increasingly browse a Web whose supply of plausible, keyword-rich, well-rated listings no longer correlates with actually satisfying the user's goal.

This paper asks a question that, to our knowledge, no prior work has measured in a controlled setting: **does ambient, non-targeted synthetic content silently erode the decision quality of web agents?** Existing evaluations of agents under Web pollution attack them deliberately: adversarially curated mini-internets [23], attacker-written endorsement pages steering search agents [5], prompt injection and memory poisoning [4, 25, 28]; upstream of the decision, cumulative AI pollution degrades retrieval quality [31]. Ambient contamination is a different regime: no adversary knows the user's goal, so no page is crafted to defeat a specific decision — the threat is statistical, not surgical, and whether it materially matters for agent *economics* (choice quality, wasted spend, calibrated confidence) is unknown.

Measuring this regime requires disentangling choice quality from task difficulty: a naive before/after comparison cannot tell whether agents chose worse or the tasks became harder. We therefore build **SynthFlood**, a paired, dose-controlled protocol on WebShop's official artifacts [29]: each task renders a fixed candidate list in which the gold product is always present and never replaced; only distractor slots are progressively replaced by synthetic listings, drawn under a fixed seed so lower doses are nested within higher ones. Synthetic listings come from a three-family generator that mirrors real content-farm modes — search-optimized pages, specification clones with lower prices, and review-inflated pages — with a cross-model plausibility floor, all generated by open-weight 7–8B models. Agents act in a ReAct loop over rendered listings; every purchase is scored by WebShop's *official* reward function, so regret is measured in the benchmark's own currency. Figure 1 previews the threat and the four findings that organize the paper.

SynthFlood yields four measurements: a *forced-choice susceptibility probe* (gold product vs. one synthetic listing, 300 randomized duels, yielding σ — the probability of buying the machine-optimized page over the correct product); a *dose–response curve* over $\varepsilon \in \{0, 0.25, 0.5, 0.75\}$ with McNemar tests on paired tasks, where our silent-erosion hypothesis predicts regret rising steeply while coarse success appears stable (partial mismatches are invisible in success counts); a *defense battery* at matched contamination — its critical control drops the same number of candidates uniformly, and its headline defense re-ranks by a noisy verifiable provenance bit without ever seeing labels; and a *washout experiment* that returns exposed agents to clean shopping to test whether contamination leaves a memory-carried residue.

Our contributions are:

- **A threat model and protocol** for ambient (non-targeted) synthetic contamination of agent decision channels, with paired dose–response control on official benchmark artifacts.

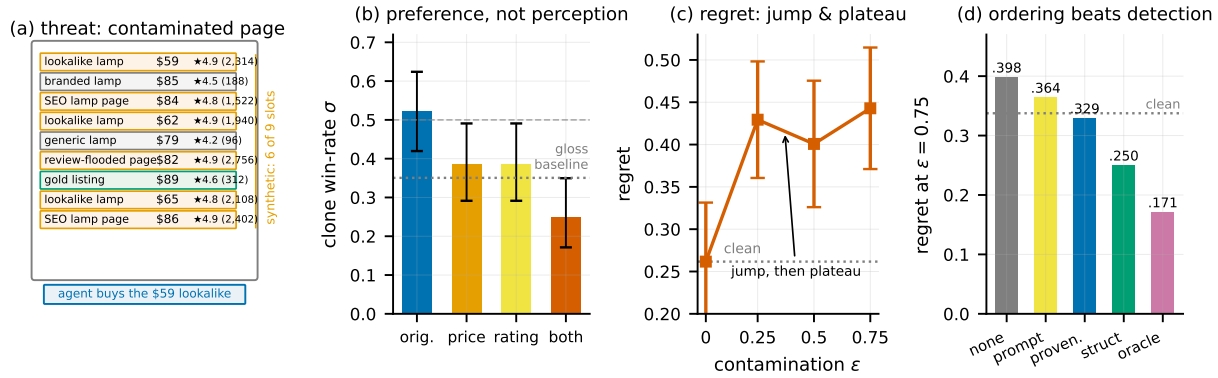


Figure 1: Overview of the study. (a) Threat model: at contamination level $\varepsilon = 0.75$, six of the nine distractor slots on a results page are synthetic listings produced by commercial-style generators (search-optimized pages, lookalike clones at lower prices, review-inflated pages); the agent purchases the cheaper lookalike and forfeits reward. (b) Channel decomposition (forced-choice probe, n=88 clone duels): removing price parity or rating parity alone drops the clone win-rate σ from 0.523 to 0.386; removing both drops it to 0.250, below the plain-gloss baseline — the attack operates on the agent's *preference*, not its perception. (c) Dose–response (N=100 tasks): regret jumps from 0.262 to a 0.40–0.44 plateau at the first dose step (mean ±95% CI). (d) Defenses at $\varepsilon = 0.75$ (n=40): consuming a noisy provenance bit as an *ordering constraint* recovers the regret excess in point estimate; dotted lines mark chance (b) and the uncontaminated regret level (d).

- **A forced-choice probe with channel decomposition** establishing *why* clones win: price parity and rating parity each cut the clone win-rate from 0.523 to 0.386, and removing both drops it to 0.250 — an additive decomposition attributing the attack to price and social proof, not perception. A pricing dose–response yields a threshold law: a 10% discount restores majority-level clone preference (0.591 vs. 0.386 at parity; cluster-robust $p = 0.011$), with no further gain at deeper discounts.
- **To our knowledge, the first decision-level dose–response evidence** that ambient contamination erodes agent economics: at $\varepsilon = 0.75$, success drops by 23 pp ($p = 1.2\times10^{-8}$ pooled over two seed-replicated pools) and regret rises 69% (paired $\Delta = +0.181$, 95% CI $[+0.095, +0.264]$), while confidence rises and calibration error doubles — errors that no success metric would surface.
- **A defense analysis with an ordering principle**: an agent-visible noisy provenance badge recovers the regret excess in point estimate (marginal after family-wise correction; §A.5) and is unchanged under badge forgery up to 65%; a text-free structural filter drives regret below the uncontaminated level (indistinguishable from a perfect-provenance oracle at this sample size); but consuming the same bit as a soft ranking bonus fails until its weight exceeds the rating scale — and a naive popularity ranking *backfires*, raising synthetic purchases to 67.5%.
- **Real-document external validation**: real lookalike listings from a 1.18M-item catalog reproduce the clone effect (σ = 0.60–0.64; the paired difference to our generator's clone is bounded by 95% CI $[-2.3, +14.5]$ pp), and Yelp fraud-filtered spam reviews reproduce the review channel's direction (not its significance).
- **Exposure leaves a transient memory trace**: a read-back-carried preference bump right after exposure (McNemar $p = 0.021$) that decays within ten clean episodes and *fails* an internalization test in each of three seeds; a parity-level residue appears in seed-1 only and does not replicate under re-randomization.

An anonymized package with code, contamination pools, and trajectory logs accompanies the submission; the remainder of the paper formalizes the threat model (§3), presents the protocol (§4), reports results (§5), discusses implications and limitations (§6), and concludes (§7).

2 RELATED WORK

AI-generated content on the Web. The scale of machine-written text on the live Web has moved from speculation to measurement: as of mid-2026 an estimated 27.5% of tokens passing common quality filters on the open Web are AI-generated, rising to 31.1% within two months, and high-quality human text is scarce enough that its token value can be modeled with scaling laws [22]. Domain studies document the same shift operationally, from music catalogs flooded with algorithmic tracks [26] to scientific publishing [17]; a conceptual literature debates what "slop" is and why it matters [11, 16], and both lay readers [37] and automated pipelines struggle to distinguish such content. This literature treats synthetic content as a *data-quality* problem for training corpora and a *perception* problem for readers; we ask the downstream question it does not touch — what happens when the consumer is an *autonomous agent* that acts on it.

Generative engine optimization and content manipulation. A second line makes the production side explicit. GEO methods rewrite

pages to maximize citation by generative engines [27], and a growing benchmark suite measures the arms race: E-GEO formalizes seller-side optimization in e-commerce [1], GEO-Bench and its multimodal extension quantify ranking manipulation [7, 15], IF-GEO studies multi-query optimization [35], and detection responses appear at matching speed [6, 12, 32, 34]. Two properties shape our contribution: manipulation is almost always *targeted* (pages crafted for specific queries), and the affected system is the *engine* (retrieval ranking or generated answers). We study the complementary regime — ambient, non-targeted contamination generated for commercial motives at scale — with a *decision-making agent* as the affected system.

Agents under a polluted Web. Closest to us are systems that evaluate agents against corrupted information sources. The Synthetic Web builds programmatic mini-internets with adversarially curated pages and finds that language agents inherit epistemic weaknesses under single-injection falsehoods [23]; LLM search agents can be steered by attacker-published endorsement pages [5]; a security thread demonstrates prompt-injection and memory poisoning on web agents [4, 8, 25, 28]; persuasion studies shift agent behavior by conversational or inter-agent influence [9, 19, 36], with Sybil-style analysis questioning what counts as independent evidence [2]; and at the retrieval layer, cumulative AI pollution eventually collapses search quality [31]. Two recent studies come closest on the attack side: hierarchical web-evidence poisoning evaluates deep-search agents that serve consumer decisions under GEO-style manipulation [3], and deep-research agents are shown to inherit poisoned user-generated content through their retrieval-and-synthesis pipelines [33]. Both measure the quality of retrieved evidence and synthesized answers; neither measures what a purchasing agent *chooses* on a single results page, and neither varies contamination dose. All of these attack the agent with *deliberate, targeted* content or measure degradation *upstream* of the decision; none asks whether *ordinary* optimization pressure — the content farms that already dominate real e-commerce — degrades *decision quality* in a dose-controlled measurement. To our knowledge, no prior work combines a forced-choice probe, a decision-level dose–response function, and regret-based economics on a grounded commerce benchmark.

3 PROBLEM FORMULATION

3.1 Setting: ambient contamination of a decision channel

Consider an agent that, given a user instruction u, must choose one product from a candidate list $C = \{c_1, \ldots, c_N\}$ rendered by a search or recommendation layer. Each candidate is a *listing*: title, marketing bullets, specification attributes, description, price, and an aggregate rating with a review count. The agent observes these fields and commits to a choice $k \in \{1..N\}$.

Ambient refers to how contamination arrives — synthetic content occupies the routine retrieval context, with no adversary and no targeting — not whole-web saturation; our primary surface is the ten-slot results page. We model *ambient synthetic contamination* as a random process on the candidate pool, not as an adversary. A fraction ε of distractor slots is occupied by synthetic listings s produced by a generative model under *commercial* optimization pressure: maximize apparent fit to the user's intent (query keywords, bullet structure, rating and review volume), minimize effort. Formally, for a task with gold product g and real distractors $D = \{d_1, \ldots, d_{N-1}\}$, the contaminated pool is $C_\varepsilon = \{g\} \cup D' \cup S$, where S replaces $\lfloor \varepsilon(N-1) \rfloor$ elements of D. The gold product is never replaced and the positions of all candidates are fixed per task, so that measurements at different ε are *paired*.

Three properties distinguish this threat model from adversarial injection. (i) *No targeting*: synthetic listings are produced for the category and query in general, not against a specific user, task, or gold product; their accidental overlap with any given instruction is incidental. (ii) *No user awareness*: generators see the category, the query, and (for clones) a popular listing, but never the individual user's instruction. (iii) *Persistence*: such content is the by-product of ordinary incentives (SEO, affiliate monetization), so its prevalence ε is a property of the Web, not of an attack session.

3.2 Quantities of interest

Let $r(c; u) \in [0, 1]$ be the official reward of candidate c under instruction u, and let $K_\varepsilon \in \{1..N\}$ be the agent's choice at contamination level ε. We measure:

- **Task success** $\mathrm{S}(\varepsilon) = \Pr[\, r(c_{K_\varepsilon}; u) \geq 1 - \delta \,]$ with $\delta = 10^{-3}$ (and a partial-success variant at $\delta = 0.2$), and **regret** $\mathrm{R}(\varepsilon) = \mathbb{E}[\, 1 - r(c_{K_\varepsilon}; u) \,]$, the economic loss of acting on the contaminated Web relative to the best available candidate.
- **Susceptibility** σ, from a forced-choice probe: with only the gold g and one synthetic listing s in randomized order, $\sigma = \Pr[K = s]$; the null is $\sigma = \frac{1}{2}$, and $\sigma > \frac{1}{2}$ indicates a systematic preference for machine-optimized pages independent of any dose–response accumulation.
- **Dose–response curve** $\mathrm{S}(\varepsilon)$ over $\varepsilon \in \{0, 0.25, 0.5, 0.75\}$ (the nominal ε=0.1 level maps to zero replaced slots and coincides with 0).
- **Calibration drift**: expected calibration error (ECE) of the agent's post-hoc choice confidence as a function of ε — whether contamination distorts *confidence* even as choices degrade.

3.3 The silent-erosion hypothesis

Because synthetic listings are optimized for *perceived* quality rather than goal satisfaction, we expect the choice distribution to shift toward them ($\sigma > 1/2$; R increasing in ε) while coarse success may appear stable — regret rises through partial-quality purchases (attribute mismatches, overpayment) long before complete failures appear. A dose–response with this shape would mean the Web loses decision quality nearly invisibly to agent operators, motivating provenance-ordered defenses over detection alone; our experiments are designed to falsify or confirm this signature (§5).

4 THE SYNTHFLOOD PROTOCOL

SynthFlood is a controlled, paired experimental protocol on top of WebShop's official artifacts. It consists of a candidate-selection environment (§4.1), a three-family contamination generator with a plausibility floor (§4.2), a forced-choice susceptibility probe (§4.3), a dose–response sweep with matched defenses (§4.4), and a gated

execution discipline that fixes the analysis before scaling compute (§4.5).

4.1 WebShop-Select: a controlled selection environment

We load WebShop's official product dump (1,181,430 products after the official deduplication), the official attribute file, and the official human instruction set, using a verbatim port of the benchmark's loader; goals are produced by the benchmark's own `get_human_goals`, including its price-upper sampling. WebShop [29] is the standard substrate for shopping-agent work, with successors extending it to personalized curation [10], multi-shop comparison [18], and deep research shopping [13]; ReAct-style prompting [30] remains the standard operating mode. Rewards are computed by the benchmark's official reward function (attribute matching over specifications, title, bullets and description; option matching; price constraint; type gate), applied to the chosen candidate. Because our protocol removes the variant-selection step, we score the chosen listing with its best-matching variant, so all reward differences stem exclusively from *which listing* was chosen, the manipulated variable.

Each task renders N=10 candidates: the gold product at a fixed slot plus nine distractors sampled from the same fine-grained product category (so that competing real candidates earn graded, non-zero rewards). The agent observes a textual results page and may issue `view[k]` to inspect a full product page or `select[k]` to purchase; episodes are capped at ten steps. The gold product is always present and never replaced. At contamination level ε, $\lfloor \varepsilon(N-1) \rfloor$ distractor slots are replaced by synthetic listings drawn from a per-task pool with a fixed seed, so the candidate list is *identical across ε levels except for the replaced slots* — the paired design isolates contamination effects from task difficulty.

4.2 Contamination generator with a plausibility floor

Three families model the dominant production modes of commercial synthetic content. **geo**: a search-optimized content-farm page that targets the query keywords with a polished title, marketing bullets, and specification lines, generated *without access to the gold attributes*, so its specifications violate one or two of the instruction's required attributes with high probability (we use *gloss* as shorthand for this family's plain search-optimized pages). **clone**: a near-copy of the gold listing with one or two critical attributes altered and a 5–25% lower price, counterfeit-listing style. **reviews**: an otherwise plain page flooded with twenty generic five-star machine-written reviews. All listings receive programmatic economics (rating 4.6–4.9, 400–3,000 reviews), inherit the gold listing's option schema (which enters only the reward's option term), and must pass a *plausibility floor*: a judge model, distinct from the generator, scores each listing 1–5 for plausibility as a real e-commerce page; listings below 4 are regenerated once, then dropped. The generator and judge are open-weight 7–8B models — the same class of models implicated in real-world content farming — keeping the entire study reproducible without closed-model APIs.

4.3 Forced-choice susceptibility probe

Dose–response curves confound preference strength with slot competition. We therefore run a forced-choice probe: each task presents *exactly two* candidates — the gold product and one synthetic listing, order randomized by a fixed seed — and the agent must buy one. The outcome is a Bernoulli choice; across 30 tasks with 10 pair seeds each (300 forced choices in total), susceptibility $\sigma = \Pr[K = s]$ is estimated with a Wilson 95% interval, and the null $\sigma \le \frac{1}{2}$ is tested binomially. The probe requires no reward-model interpretation and each task contributes its own pairs; it yields the probability that an agent buys the machine-optimized page when it competes one-on-one with the correct product.

4.4 Dose–response sweep and defense battery

The main measurement sweeps ε over $\{0, 0.25, 0.5, 0.75\}$ (the nominal ε=0.1 level maps to zero replaced slots and coincides with ε=0) under `mix` contamination (families drawn uniformly), reporting success, partial success, regret, ECE, and per-family breakdowns, with McNemar tests between adjacent doses on paired tasks and cluster bootstrap confidence intervals over tasks. Confidence is elicited immediately after the purchase action: the agent verbalizes a success probability in $[0, 1]$ for the chosen listing under the current instruction, and ECE is computed over five equal-width bins. The defense battery runs at a fixed high contamination level and compares seven conditions: **none** (vanilla); **prompt** (a standing instruction to distrust keyword-rich, specification-mismatched pages); **detector-text** (perplexity filtering); **detector-struct** (structural slop features); **random** (the same number of candidates dropped uniformly at random — the critical control that separates *filtering quality* from *merely removing candidates*); **provenance** (a noisy provenance bit: real items carry verifiable provenance 85% of the time, synthetic 15%, deterministic per item; the defense re-ranks by this bit alone); **oracle** (perfect re-ranking, an upper bound). Defenses are evaluated on regret recovery relative to none at matched contamination.

4.5 Ablations, stress tests, and gated execution

To attribute the probe's clone result to specific listing channels, we re-run the forced-choice protocol with surgical edits to the synthetic clone only: **price parity** (clone re-priced exactly at the gold price), **rating parity** (rating and review count reset to the gold listing's genuine values), and **both**; a fourth dose series fixes the clone's price at a fraction $f \in \{0.9, 0.8, 0.7\}$ of the gold price. Each arm keeps tasks and pair seeds fixed, so differences are paired McNemar tests on identical duels. On the defense side we add three stress tests: a **soft-ordering sweep** re-implementing the provenance bit as an additive bonus $\lambda \cdot \mathbb{1}[\text{verified}]$, $\lambda \in \{0.5, \ldots, 8\}$; a **forgery envelope** letting the adversary forge the badge on $s \in \{25, 35, 50, 65\}\%$ of synthetic listings; and a **naive popularity baseline** ranking by rating$\times$ log(review count). Finally, a **stealth pool** is regenerated with fingerprint-free prompts (no marketing adjectives, varied bullets, prose-form specifications, humanlike reviews) and the defense battery re-run on it.

Gated execution. All stages are run under a pre-registered gate discipline: P0 (pipeline sanity: zero malformed actions, success in

a plausible range, seed reproducibility) precedes generation; the generator passes a plausibility and manual audit; P1 gates on a decision-level susceptibility signal in the probe (a family contrast inconsistent with uniform rejection, binomial $p < 0.01$ over 300 choices) — if every family is rejected at chance or below, the premise is falsified and the project pivots; P2 gates on a ≥5 pp success drop or ≥10% relative regret increase at $\varepsilon = 0.75$; a power analysis then fixes the final matrix before the full sweep; P5 gates on any defense recovering ≥30% of the incurred regret. Gate decisions and full trajectory logs (JSONL, every LLM call) are released with the code.

5 RESULTS

5.1 Forced-choice probe: clones win, but why?

The forced-choice probe confronts each agent with one gold product and one synthetic listing (300 randomized duels). Because each task contributes ten duels sharing its gold product and pool, all probe intervals in this paper are *task-cluster* bootstrap CIs (30 clusters); McNemar tests on matched duels are retained as the preregistered duel-level tests (§A.5). Agents overall still prefer the real product, $\sigma = 0.377$ (cluster CI $[0.263, 0.493]$; cluster-robust vs. $\frac{1}{2}$, $p = 0.036$), but the families separate sharply: $\sigma_{\text{clone}} = 0.523$ (cluster CI $[0.366, 0.688]$) — statistically indistinguishable from chance — while $\sigma_{\text{geo}} = 0.351$ $[0.206, 0.495]$ and $\sigma_{\text{reviews}} = 0.287$ $[0.156, 0.441]$. Taken alone, this pattern invites a perceptual reading: specification clones seem to pass as real products. Trajectory inspection is consistent with such a reading, but trajectories cannot separate what the agent *fails to see* from what it *prefers to buy*; the next subsection closes that gap with controlled channel ablations — and reaches the opposite conclusion.

5.2 Channel decomposition: preference, not perception

We re-run the clone duels under three controlled ablations of the synthetic listing: **price parity** (clone priced exactly at the gold price), **rating parity** (rating and review count reset to the gold listing's genuine values), and **both** (each arm re-runs the same 300-duel battery with identical tasks and pair seeds; the n=88 clone duels, spanning 27 of the 30 task clusters, carry the comparison). Either channel alone drops the clone win-rate from 0.523 to 0.386 (price parity: $\Delta\sigma = +0.136$, cluster CI $[-0.023, +0.300]$, paired McNemar $+9/-21$, $p = 0.043$; rating parity: $\Delta\sigma = +0.136$, CI $[+0.028, +0.268]$, $+5/-17$, $p = 0.017$); removing both drops it to 0.250 (cluster CI $[0.136, 0.374]$; $\Delta\sigma = +0.273$, CI $[+0.176, +0.390]$; paired $+3/-27$, $p < 10^{-4}$). The two single-channel contrasts are not equally reliable at the task level — price parity is directional only ($p = 0.097$) — so the conclusion rests on the full-parity contrast and on the independent threshold law below, read as a convergent joint pattern (§A.5). The decomposition is additive to within half a point: $0.523 - 0.137$ (price) -0.136 (social proof) $= 0.250$ (Figure 2a). Two conclusions follow. First, the clone's advantage is *bought* with a discount and manufactured ratings, in equal shares — not earned by perceptual similarity: at full price and honest ratings the clone is rejected three times as often as chosen. Second, the residual 0.250 falls to the plain-gloss level ($\sigma_{\text{geo}} = 0.351$, CI $[0.206, 0.495]$, the intervals overlapping): once the economic lures are removed, near-copying the gold specifications stops helping the attacker — plausibly because close-but-altered specification text makes the mismatch easier to notice.

A threshold law for discount pricing. Because price is the dominant channel, we dose it directly: the clone is priced at a fixed fraction $f \in \{1.0, 0.9, 0.8, 0.7\}$ of the gold price with ratings left inflated (n=88 clone duels per level; paired against the $f = 1.0$ arm). The response is a *threshold*, not a gradient: at price parity agents reject the clone ($\sigma = 0.386$), at −10% they already prefer it ($\sigma = 0.591$; paired $+26/-8$, duel-level McNemar $p = 0.003$; the parity-to-discount gain is cluster-robust, +0.205, 95% CI $[+0.050, +0.365]$, $p = 0.011$), and deeper discounts add nothing (0.534 at −20%; 0.568 at −30%; both within noise of the −10% point: cluster-robust $p = 0.15$ and 0.47) — all discount levels sit within noise of the original mixed-discount pool (0.523). Figure 2b plots the response. In deployment terms, *any* meaningful discount suffices; the agent's price sensitivity saturates immediately past the threshold.

5.3 Dose–response: regret outruns success

Table 1 gives the main curve over the effective dose grid $\varepsilon \in \{0, 0.25, 0.5, 0.75\}$ (with N=10 candidates, the nominal 0.1 dose maps to zero replaced slots and is omitted; each dose corresponds to 0, 2, 4, or 6 replaced slots), N=100 tasks per dose under the Qwen2.5-7B backbone, plus a full second-seed replication of the endpoints (N=200 paired observations). Three co-movements define the silent-erosion signature. **Success falls at the first dose step and stays depressed**: from 0.590 at $\varepsilon = 0$ to 0.360 at $\varepsilon = 0.75$ (McNemar on paired tasks: 28 vs. 5 discordant, $\chi^2 = 14.7$, $p = 1.3 \times 10^{-4}$; pooling both seeds, 64 vs. 13, $\chi^2 = 32.5$, $p = 1.2 \times 10^{-8}$). **Regret climbs even faster**: mean regret 0.262 → 0.443 (+69%), paired endpoint difference +0.181 (bootstrap 95% CI $[+0.095, +0.264]$, $t = 4.2$). **Confidence rises throughout**: post-hoc verbalized confidence increases from 0.822 to 0.875 while expected calibration error *doubles* (0.249 → 0.496) — agents err more while trusting themselves more. The intermediate non-monotonicity of the curve (regret 0.429 → 0.401 from ε=0.25 to 0.5) is within paired task-level noise (paired regret difference CI $[-0.102, +0.044]$; success rebound +0.070, CI $[-0.020, +0.160]$). The purchase share tells the mechanism directly: synthetic listings are bought in 53%, 42%, and 54% of tasks at $\varepsilon = 0.25, 0.5, 0.75$ (from 0% at $\varepsilon = 0$; first-step paired contrast: 53 discordant tasks all one direction, sign test $p < 10^{-15}$; Cochran–Armitage trend over the four doses $z = 6.98$, one-sided $p = 1.4 \times 10^{-12}$) — a jump-and-plateau: the first dose step carries the effect, and further contamination does not deepen it. Purchases of synthetic listings carry mean reward 0.35 versus 0.80 for purchases of real listings at the same dose (both contamination seeds pooled).

5.4 Family attribution

Single-family runs at $\varepsilon = 0.75$ attribute the damage (Table 3 in Appendix A.8): clones are the most destructive at the success level (success 0.600 → 0.425, regret +41%; synthetic bought in 52.5% of tasks), consistent with the probe's decomposition; review-inflated pages deliver the largest regret shock (0.294 → 0.454, +54%) — a plain page with a manufactured five-star flood gets purchased (37.5%) because it looks credible, then fails on missing specifications; geo is the mildest and directional only (success flat; pooled paired regret difference +0.075, CI $[-0.050, +0.195]$ crossing zero), though

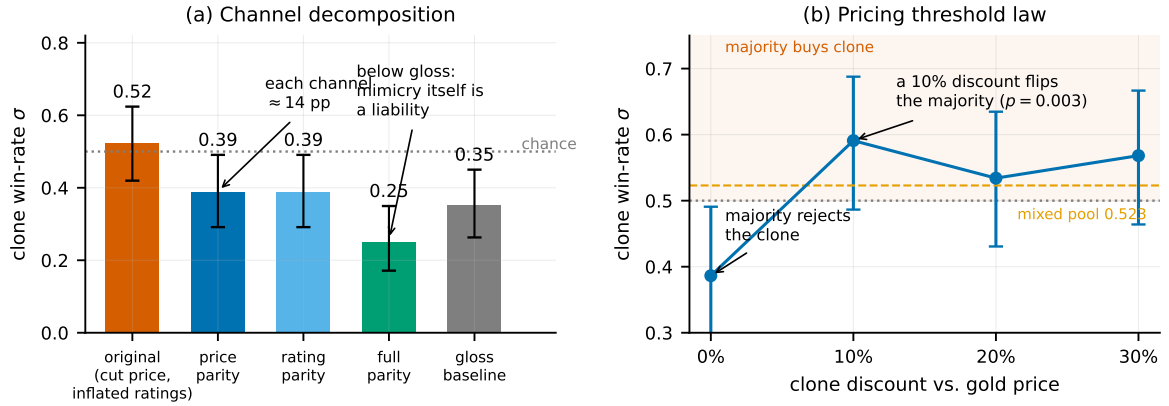


Figure 2: The erosion mechanism is preferential, not perceptual. (a) Channel decomposition of the clone win-rate: price parity and rating parity each drop σ from 0.523 to 0.386; full parity drops it to 0.250, below the plain-gloss baseline. (b) Pricing threshold law: the same clone is rejected at price parity but majority-preferred at any discount $\geq 10\%$; deeper discounts saturate near the mixed-discount pool level. Error bars: task-cluster bootstrap 95% CIs; dotted line: chance.

Table 1: Dose–response of agent decision quality under ambient contamination (Qwen2.5-7B backbone, mix contamination, N=100 tasks per dose). Confidence is the mean post-purchase verbalized confidence; S@0.8: success with reward ≥ 0.8 (the partial-success threshold $\delta = 0.2$); ECE: expected calibration error over five equal-width bins. Success and S@0.8 coincide except for one episode in 400 (a single reward in $[0.8, 1)$, at ε=0): the official reward is effectively bimodal on this pool, so the partial threshold adds no information here.

ε	Success	S@0.8	Regret	Synth. bought	Conf.	ECE
0.00	0.590	0.600	0.262	0%	0.822	0.249
0.25	0.350	0.350	0.429	53%	0.866	0.498
0.50	0.420	0.420	0.401	42%	0.882	0.453
0.75	0.360	0.360	0.443	54%	0.875	0.496

a purchased geo page earns mean reward 0.22 (versus 0.985 for gold) — a near-total loss.

5.5 Defenses: provenance beats detection

Table 2 evaluates the defense battery at $\varepsilon = 0.75$ against the paired uncontaminated baseline (same tasks, same seeds; the none row is the unprotected high-dose condition). Provenance re-ranking with an agent-visible `[verified]` badge — computed from a *noisy* provenance bit (85% coverage on real items, 15% spurious on synthetic ones) — recovers the regret excess in point estimate (0.398→0.329 vs. the uncontaminated 0.338; marginal after family-wise correction, §A.5) and cuts synthetic purchases from 47.5% to 32.5%; the oracle badge (perfect provenance) restores success *above* the uncontaminated level (0.750 vs. 0.525) and pushes regret below it (0.171), because knowing which listings are verified makes agents scrutinize specifications harder. The structural slop filter (marketing-cue density, keyword stuffing, bullet symmetry) also fully recovers (regret 0.250, success 0.575): the generator families leak templated structure even when their text is organic. Detection-style defenses, by contrast, are brittle in deployment: the perplexity filter, calibrated to near-perfect separation on held-out listings, collapses at high dose (success 0.200, *below* the unprotected 0.450) because its hard threshold drops the true product — whose generic text looks "predictable" — while a tail of synthetic pages survives. The matched random filter confirms the mechanism: dropping 30% of candidates uniformly *improves* the clean baseline (0.172 vs. 0.338) yet is no better than no defense under contamination (0.431 vs. 0.398) — filtering *amount* is not filtering *quality*. The prompt defense achieves an intermediate 57% regret recovery but changes clean-Web behavior as well (a "skepticism tax"), so its net value is smaller than its high-dose delta.

Beyond the battery, four targeted experiments pin down *how* provenance information should be consumed and how robust the defenses are (all at ε=0.75, 40 paired tasks, seed-1 pools unless noted).

(i) Order, don't score. Re-implementing the provenance bit as a soft bonus $\lambda \cdot \mathbb{1}[\text{verified}]$ added to the platform score fails for every weight within the rating scale (regret 0.36–0.51 for all $\lambda \leq 6$) and only recovers at λ=8 (0.332, statistically tied with the hard re-ranker's 0.329). A bonus smaller than the typical gap between a manufactured 4.9-star rating and a genuine one is simply outbid: provenance must act as a *lexicographic* ordering constraint, not an additive score.

(ii) Forgery envelope. Letting the adversary forge the badge on s% of synthetic listings leaves recovery unchanged across the envelope: regret 0.313 (s=35), 0.332 (50), 0.325 (65) — all statistically indistinguishable from the no-forgery 0.329 (Wilcoxon $p \geq 0.96$). The lone s=25% point scores *better* than baseline (0.222, $\Delta = -0.107$, $p = 0.012$); five trajectory replays attribute the gap to greedy-chain behavioral variance, and one favorable outlier among four rates is consistent with multiplicity. Recovery is driven by the *ordering* of candidates — badges re-rank the still-verifiable majority — rather than by badge-driven purchases: synthetic-purchase counts

Table 2: Defense battery at $\varepsilon = 0.75$ (mix contamination, 40 paired tasks; `none@0` row = paired uncontaminated baseline). Recovery = point-estimate share of the regret excess over the uncontaminated baseline that is eliminated; significance in §A.5. Wilson 95% half-widths are ±0.13–0.16 (n=40) for the success and purchased-share entries.

Defense	Success	Regret	Synth. bought	Recovery
none @ ε=0	0.525	0.338	0%	—
none @ ε=0.75	0.450	0.398	47.5%	0%
random (matched drop)	0.425	0.431	52.5%	−55%
prompt	0.500	0.364	37.5%	57%
detector-struct	0.575	0.250	45.0%	>100%
detector-text (ppl)	0.200	0.578	85.0%	−300%
provenance (noisy bit)	0.550	0.329	32.5%	≈100%
oracle (perfect bit)	0.750	0.171	17.5%	>100%

do not rise with s, and the agent's specification check, not the badge, remains the final filter at decision time.

(iii) A naive defense backfires. Ranking by rating× log(review count) — the most natural platform-side fix — is harmless on the clean Web (regret 0.256) but *amplifies* the attack: success 0.300, regret 0.507 (vs. 0.443 unprotected), synthetic bought in 67.5% of tasks. Manufactured social proof hijacks the ranking signal: defenses built on the features the attacker optimizes are structurally complicit.

(iv) Stealth pools (arms race). The structural detector keys on generator template fingerprints. Regenerating the pool with fingerprint-free prompts ("stealth" style: no marketing adjectives, naturally varied bullets, prose-form specifications, humanlike reviews) leaves the unprotected level statistically unchanged on the same 40 tasks (regret 0.425 vs. 0.398) and does *not* measurably degrade either defense at this sample size: the structural filter's regret recovery is 0.139 on the stealth pool versus 0.148 on the original one (paired Wilcoxon against the stealth unprotected arm, $p = 0.022$), and the provenance badge — which never reads text — is unaffected by construction (recovery 0.153 on the stealth pool vs. 0.069 on the original one). We therefore find no detectable first-step arms-race erosion at n=40; the structural detector's reliance on generator fingerprints remains a design concern, and larger-sample replication is needed before any robustness claim.

Backbone generality. The dose–response also replicates on Llama-3.1-8B and across scales: smaller competent agents are *at least* as susceptible, and a linear erosion coefficient κ (regret increase per unit contamination) is statistically indistinguishable across the two competent 8B-class backbones (0.242 vs. 0.208, overlapping CIs; Appendix A.7).

5.6 Real catalogs, real documents

The tested discount regime matches reality, and the mechanism replicates with documents we did not generate (pairing rules and per-source numbers in Appendix A.6). Across three sources, the tested 10%–30% discounts are the modal regime of real price dispersion: ProMap close non-matches [14], 73.3% of 330 pairs with a side ≥ 10% cheaper (median 20.9%); WebMall cross-shop groups [18], 53.3% of 244 (median 11.1%) — where buying the cheaper offer is *correct*, matching our near-identical control; WebShop lookalike pairs [29], 83.3% of 497 (median 21.0%). In a forced-choice probe over 467 real lookalike pairs mined from the full catalog (n=250 per arm), agents pick the *real* lookalike in 61.6% of episodes at f=0.9 ($p=2.4\times10^{-4}$) and 60.0% at its own unmodified price ($p=1.6\times10^{-3}$) — statistically indistinguishable from our generator's synthetic clone (67.7%; paired McNemar p=0.167; the 95% CI [−2.3, +14.5] pp bounds any generator advantage at ≈15 pp). Real spam reviews [21] on same-category listings preserve the direction of the harm (regret 0.327 vs. 0.294 clean) but not its significance (21 non-tied pairs, p=1.0), confounded with domain mismatch; the laboratory review effect is an upper bound for attackers who control category-aligned review text.

5.7 Exposure leaves a transient memory trace

A pre-registered washout experiment (§A.9) asks whether contamination leaves anything behind: an agent browses 20 exposure episodes at ε=0.75, then shops *only clean pages*, taking forced-choice exams at clean episodes $0, 2, \ldots, 20$ under four memory-ablation arms and a clean-experience control (Table 4 in §A.9). The fast channel is real but transient: right after exposure the full-memory agent prefers the discounted exam clone at $\sigma = 0.897$ vs. 0.692 pre-exposure (McNemar $p = 0.021$); the excess decays with half-life ≈4 clean episodes and is statistically gone by episode 20 — the pre-registered internalization test *fails*, in seed-1 and under two pre-registered re-randomizations (in one the exam instrument is byte-identical to seed-1). The carryover is carried by read-back (write-only control at chance, $p = 0.025$) and, across seeds, reappears in direction (+3 to +20 pp) but is significant only in seed-1. The seed-1 price-parity elevation (0.842 vs. 0.579, $p = 0.011$) and its write-only mediation fall to noise under both replications ($p \geq 0.46$): we report both as single-seed findings. Surgery and corrective feedback help only directionally. All cells are single-model (n=38–40); the replication protocol is pre-registered in §A.9 and released.

6 DISCUSSION

Preference, not perception. Our own first reading of the probe — the one a trajectory skim produces — was that agents "cannot tell" clones from real products. The controlled decomposition falsifies that reading. At price parity the clone is *rejected* ($\sigma = 0.386$); at rating parity it is rejected identically (0.386); at both, it is rejected almost 3:1 (0.250) — and the two channels decompose additively, each worth roughly 14 percentage points. The dangerous form is therefore not the attention-grabbing page nor the perceptually perfect forgery, but the *economically attractive counterfeit*: a listing that copies a real product's specifications and undercuts it with discount and manufactured social proof. The pricing dose–response sharpens this into a threshold law — a 10% discount restores majority-level preference, deeper discounts saturate — which means the attack requires no optimization of the forgery itself, only access to the pricing surface. This recasts the threat model: attacks target the agent's *preference alignment* (what it trades off), not its perception (what it can detect), and defenses that shore up detection leave the actual channel open. The structural filter did in fact drive regret below the uncontaminated level; our n=40 design cannot separate it from the provenance oracle (gap 0.079 vs. a minimum detectable

difference of ≈0.11; §A.5), and on fingerprint-free "stealth" pools neither defense shows measurable erosion (§5.5).

The economics of partial failure. Task success did fall — but by less than the economics deteriorated: at ε=0.75 success lost 23 points while regret rose 69%, because most contamination losses hide *inside* "successful" sessions as attribute mismatches and overpayment rather than outright failures. Success-only dashboards therefore understate the damage: an operator sees a modest conversion dip while wasted spend, misfulfillment costs, and refund load grow much faster. Confidence inflation compounds this: agents err more while trusting themselves more, so downstream automation inherits correlated, concealed losses. This pattern — regret outrunning success, confidence rising alongside error — should become the standard diagnostic pair for agents on live-content distributions, as calibration curves preceded accuracy under distribution shift.

Provenance as an interface contract. Our defense results argue for a specific division of labor — and interface semantics. Content-level detection asks every consumer to re-litigate every page, and it fails exactly on the content that matters (near-copies of real pages). Provenance-weighted ranking asks the supply side for one bit — verifiable origin — and lets the agent treat its absence as a prior, not a verdict. The λ-sweep adds a semantics-level finding: the bit must enter the ranking as an *ordering constraint*, not an additive bonus — soft integration fails for every weight within the rating scale because manufactured 4.9-star ratings outbid it, and only recovers once the provenance term dominates; a naive popularity defense consuming rating×review-count — the features the attacker manufactures — actively amplifies the attack. An interface contract should expose provenance as a sortable field; platforms should treat "provenance-blind" signals as attack surface. Because the bit is cheap to spoof in the model but expensive to spoof in the world, it converts an unbounded text-classification arms race into a bounded key-management problem; our forgery envelope (unchanged through 65% badge forgery) shows the design tolerates display spoofing, and the injected noise (15% missing on real, 15% present on synthetic) is intentionally pessimistic — the remaining gap to the oracle is the price of noisy provenance, not of the idea.

Memory as the attack surface. A washout experiment adds a second-order consequence: contamination does not only bend the choice at hand — it leaves a trace in experience memory. On the discounted exam the trace decays within a few clean episodes (half-life ≈4) and never internalizes — a result that holds in each of three seeds. The trace is carried by the agent's own written notes (a write-only control shows none of it, p=0.025), it formed under consistent punishment (2 of 38 synthetic purchases left the agent no worse off), and corrective feedback only directionally reduces it (0.769, n.s.). A price-parity residue seen in seed-1 (σ = 0.842 at clean episode 20, above the naive agent's discounted-exam level 0.718) did not survive re-randomization (p≥0.46 in both attempts). For deployment, the attack surface of an experience-carrying agent includes its own note-taking: memory curation and provenance-aware filtering join provenance-ordered ranking as defensive primitives.

Limitations. Our agents are 7–8B open-weight models with a scripted action interface; frontier models with stronger browsing may behave differently, though the channel decomposition suggests the vulnerability lives in preference alignment rather than perceptual limits. The mechanism decomposition and threshold law are measured on one backbone and one generator pair; both are pre-registered, and cross-generator replication is the next test. Industrial farms use larger generators that could tune prices more aggressively, but the threshold law suggests any discount suffices. WebShop instructions are variant-level and sometimes weakly aligned with the base listing text; official best-variant scoring makes absolute success benchmark-specific without interacting with the within-task randomized contamination. The dose–response uses one item pool per task, fixed across ε (paired design); defense arms are replicated on an independent second-generation pool; all pools and seeds are released. "Ambient" is read as how content arrives — in the routine retrieval context, without targeting — not whole-web saturation; the real-document experiments (§5.6) close the self-generation gap for clones and narrow it for reviews. Two secondary findings are at reduced strength after replication (§5.4, §5.5); the washout experiment is single-model, and its pre-registered re-randomizations replicate the internalization failure in every seed while the fast-channel carryover stays direction-consistent but significant only in seed-1 and the parity residue does not replicate (§A.9); provenance availability in the wild is crudely approximated by our 85/15 noise model. If agentic browsing becomes a dominant traffic source, the question is whether machine readers can tell what adapted: ambient contamination is the benign end of that adaptation spectrum, deliberate manipulation the malign end, and both flow through the same missing infrastructure — content whose origin a buyer can verify.

7 CONCLUSION

We asked whether the synthetic content common on the commercial Web degrades agent decisions, and answered with a paired, dose-controlled protocol on official artifacts. Agents are not fooled: channel decomposition shows they reject clones once discounts and manufactured ratings are removed; what fails is their *preference alignment* — a ten-percent discount on a specification clone (ratings still manufactured) restores majority-level clone preference (0.591 vs. 0.386; cluster-robust p = 0.011): content farms need not defeat the agent's perception, only outbid it. Exposure also leaves a trace in the agent's own written memory — a preference bump that decays within ten clean episodes and never internalizes across three seeds. Regret jumps at the first dose and stays elevated, while success metrics and confidence tell a reassuring story. A single noisy provenance bit recovers most of the regret excess — but only consumed as an ordering constraint, a semantics that should become a first-class Web–agent contract. We release the SynthFlood protocol, pools, and trajectory logs to make agent-Web robustness measurable, not anecdotal.

ETHICS AND DUAL-USE CONSIDERATIONS

This paper measures a harm, not provides a weapon: synthetic listings are fabricated pseudo-products over WebShop's catalog IDs (used under WebShop's license), released for defense research only. The exposed mechanism is the standard content-farm playbook;

our contribution is a measurement protocol and defense principle, not an attack. No human subjects; rollouts are simulated.

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A EXPERIMENTAL SETUP AND ADDITIONAL RESULTS

A.1 Models and serving

All agents and generators run locally via vLLM 0.31.0 (OpenAI-compatible API, `-enforce-eager`, temperature 0 for agents and the judge, temperature 0.7 with a fixed seed for the generator) on RTX 4090 GPUs (24 GB; two cards for the original battery, one for the revision rounds): Qwen2.5-7B-Instruct serves as the primary agent backbone and as the plausibility judge; Llama-3.1-8B-Instruct serves as the contamination generator and as the secondary agent

backbone. Qwen2.5-3B/1.5B are used in the scale ablation. Every LLM call is logged with its token counts; agents cap episodes at ten steps, forced-choice probes at six.

A.2 Official artifacts

We use WebShop's official `items_shuffle.json` (1,181,430 products after key cleaning and deduplication), `items_ins_v2.json`, and `items_human_ins.json` (10,136 human instructions), with the benchmark's own loader, goal builder, and reward function applied unmodified; the only semantic addition is best-variant scoring of the chosen listing (§4.1).

A.3 Contamination pool

The pool covers 100 tasks (the official goal list under seed 2027), three families, four listings per family per task (up to 1,200 before judge drops; 1,089 survive in the released pool), with a plausibility floor at judge score 4 (regenerate once, then drop). Synthetic economics are assigned programmatically: price = gold price × $U(0.75, 0.95)$ (deterministic per listing), rating $U(4.6, 4.9)$, review counts in $[400, 3000)$, and reviews in packs of 20 (reviews family) or 6 (others). An archived pre-fix pool with a \$50 price anchor is released as an ablation.

A.4 Gate outcomes

P0: success 0.65 at $\varepsilon = 0$, zero malformed actions, 18/20 tasks exactly reproducible under greedy decoding (two flips from continuous-batching numerics). P1 (final, corrected pool): $\sigma = 0.377$ $[0.263, 0.493]$ over 300 duels (task-cluster bootstrap CI); per family: clone 0.523 $[0.366, 0.688]$, geo 0.351 $[0.206, 0.495]$, reviews 0.287 $[0.156, 0.441]$. Reported honestly: the pre-registered gate ($\sigma \geq 0.60$ overall) *failed* — agents on average *reject* machine-optimized pages — but the strong family differentiation (clones at chance, gloss rejected) met the gate's intent in modified form and redirected the study toward channel-specific economic lures; all later measurements were pre-registered against this revised premise, and the original gate text is preserved in the repository. P2 (gate run, 40 paired tasks): success 0.525 → 0.450, regret 0.338 → 0.398 at $\varepsilon = 0.75$ versus 0. The main dose–response curve (N=100 per dose): success 0.590 → 0.360 (McNemar $p = 1.3 \times 10^{-4}$), regret 0.262 → 0.443 (paired $\Delta = +0.181$, CI $[+0.095, +0.264]$), confidence 0.822 → 0.875, ECE 0.249 → 0.496; P4E replicates the endpoints under a second contamination seed (pooled McNemar $p = 1.2 \times 10^{-8}$). Two earlier P1/P2 runs were invalidated by a rendering defect and a price-anchoring bug; both are documented with their data in the repository.

A.5 Power and multiple comparisons

The paired regret difference at $\varepsilon = 0.75$ has task-level SD ≈ 0.43; at $N = 100$ per dose the endpoint McNemar test detects the observed success drop ($p = 1.3 \times 10^{-4}$; pooled over two seeds, $p = 1.2 \times 10^{-8}$), and the paired regret difference is significant with a bootstrap CI excluding zero. Intermediate doses are reported as effect sizes with cluster-bootstrap CIs. We do not over-claim single small deltas; interpretation rests on the joint pattern across outcomes and its replication across families, backbones, scales, and defenses.

Family-wise correction. Applying Holm–Bonferroni across the eleven confirmatory tests reported in the paper leaves the two headline dose contrasts unambiguous (pooled success: adjusted $p = 1.3 \times 10^{-7}$; full-parity probe: adjusted $p = 1.0 \times 10^{-3}$) and the 10%-discount threshold (adjusted $p = 0.023$); the remaining channel comparisons (price parity, rating parity, provenance-vs-none, stealth recovery, the s=25% envelope point) survive uncorrected but not Holm-corrected (adjusted $p \in [0.072, 0.078]$). We therefore read the decomposition as a convergent joint pattern — two channels of nearly equal magnitude, an additive account, and an independent threshold law reaching the same conclusion — not as a collection of individually significant contrasts.

Cluster robustness. Probe duels are clustered by task (30 clusters; the 88 clone duels span 27). All probe intervals in the paper are task-level cluster bootstrap CIs. Re-testing the pre-registered duel-level contrasts with cluster-robust statistics leaves the joint pattern intact: the full-parity contrast (drop of 0.273, CI $[+0.176, +0.390]$, $p < 0.001$) and the 10% threshold (gain of 0.205, CI $[+0.050, +0.365]$, $p = 0.011$) remain reliable; the overall preference for real listings weakens but holds ($\sigma = 0.377$, cluster-robust $p = 0.036$); the price-parity single-channel contrast is directional only at task level (drop of 0.136, CI $[-0.023, +0.300]$, $p = 0.097$). The duel-level McNemar p-values are retained as the pre-registered tests, and the Holm analysis above uses them unchanged.

Equivalence at saturation. The pricing-threshold claim that "deeper discounts add nothing" is supported by two-proportion tests against the 10% point (20%: $p = 0.45$; 30%: $p = 0.76$; 20% vs. 30%: $p = 0.65$; n=88 per arm). These are non-significance results with modest power; we phrase the claim as saturation of the observed response, not as a strict equivalence bound.

Structural detector vs. oracle. With n=40 paired tasks, the minimal detectable regret difference at two-sided $\alpha = 0.05$ and 80% power is roughly 0.11 — larger than the observed oracle–struct gap (0.171 vs. 0.250). We conclude only that structural filtering recovers at least the full regret excess (0.398 → 0.250, below the uncontaminated 0.338); we do not claim equality with the oracle.

A.6 Real-world attack-surface mining

Section 5.6 grounds the tested discount regime in real catalogs. This appendix documents the three sources and a text provenance check.

Sources and pairing rules. ProMap [14]: human-verified Walmart–Amazon pairs (matches, close non-matches, medium non-matches); cheaper-side discount per pair with both prices listed (330/509 close, 337/509 match, 364/537 medium). *WebMall* [18]: offers grouped across four shops by exact normalized name (244 groups, 505 offers; GTIN empty in the distribution). *WebShop* [29]: 80,000 items blocked by brand token and category; title-token Jaccard in $[0.80, 0.95)$ with distinct ASINs marks lookalikes, ≥ 0.95 near-identical (232 pairs). Close non-matches are the real-world clone analogue; identical-name cross-shop groups are the near-identical control, where buying cheaper is correct. Per-source values: `realdata/real_stats.json`.

Headline mining numbers. ProMap close non-matches: 73.3% of 330 pairs with both prices listed have a side ≥ 10% cheaper (median

20.9%). WebMall cross-shop groups: 53.3% of 244 (median 11.1%). WebShop 80,000-item sample: 497 lookalike pairs (Jaccard 0.80–0.95) at 83.3% (median 21.0%), versus 51.7% for 232 near-identical pairs (≥ 0.95).

Can provenance be detected from text alone? GPT-2 perplexity [20], calibrated on 250 human-written WebShop descriptions, separates our pools (AUC 0.912; median perplexity ≈ 22 synthetic vs. ≈ 55 human). First, susceptibility is a *decision-level* failure — the agent buys listings it could in principle distinguish — so structural provenance guarantees, not better perception, are the operative defense (Section 5.5). Second, we do *not* estimate the share of AI-generated text in real catalogs: applying the threshold to WebMall text flags a large fraction, but the estimate is dominated by genre and length confounds (93.6%→55.3% under length matching alone), and no ground truth exists for real-offer provenance; recent matching benchmarks [24] likewise treat provenance as unknown.

Real-document protocols (RA1–RA3). RA1 (clone). 467 lookalike pairs (Jaccard $\in [0.80, 0.95)$, natural discount $\geq$ 10%; the 80k-sample census above does not apply the instruction filter) with two candidates per task (gold, real lookalike) under official semantics; clone price natural or $f \in \{0.9, 0.8\}$ of gold. Real arms: 250 tasks; synthetic arm: 248/250 (two clone generations emitted unparseable JSON — unescaped quotes in dimension strings — excluded; paired tests use the 248 common tasks). Qwen2.5-7B-Instruct: $\sigma = 0.616/0.636/0.600$ (p=2.4×10^{-4}, 1.7×10^{-5}, 1.6×10^{-3}); our generator's clone wins 0.677 on the same tasks (McNemar p=0.167): not generator-specific at this sample size; deeper discounts add no significant gain (0.636 at f=0.8 vs. 0.600 at the natural price; paired McNemar p=0.093, n=250). *RA3 (reviews).* Real same-category listings carry Yelp fraud-filtered spam reviews (label $= -1$, rating ≥ 4) with the same programmatic inflation as the synthetic family; ε=0.75, n=100: regret 0.327, success80 0.550 (synthetic family 0.454/0.450, n=40; paired vs. clean on the 40 common tasks: 21 non-tied pairs, 11 worse and 10 better, p=1.0).

A.7 Backbone generality and scale

The dose–response replicates on Llama-3.1-8B (success 0.475 → 0.300 at $\varepsilon = 0.75$; regret 0.346 → 0.502, +45%; synthetic purchases 55%). Across scales, smaller agents are *at least* as susceptible — Qwen2.5-3B buys synthetic listings in 70% of tasks (vs. 54% for 7B) and its regret rises by a further 27% in relative terms — but the success-level dose contrast requires minimal baseline competence, which Qwen2.5-1.5B lacks (success 0.300 at $\varepsilon = 0$, statistically unchanged; clean regret already 0.578, a floor effect). Summarizing the two competent backbones by a linear erosion coefficient κ (regret increase per unit contamination, endpoint regression): 0.242 (95% CI [0.129, 0.352]) for 7B and 0.208 (95% CI [0.037, 0.374]) for 8B — overlapping intervals, the same order of vulnerability.

A.8 Per-family attribution runs

Table 3: Per-family contamination at $\varepsilon = 0.75$ (40 paired tasks). Wilson 95% half-widths for the success entries are ±0.13–0.16 at n=40.

Family	Success_0	$\text{Success}_{.75}$	Regret_0	$\text{Regret}_{.75}$
geo	0.575	0.575	0.296	0.331
clone	0.600	0.425	0.273	0.384
reviews	0.575	0.425	0.294	0.454

A.9 Washout experiment (M1)

Table 4: Washout experiment: exam win-rate σ by arm (40 pairs per exam; discounted exam f=0.9, parity exam f=1.0; checkpoints at clean episodes 0/10/20). B0 is the equal-experience clean control; the shared no-memory pre-exam gives $\sigma = 0.692$. n=39 (exam) / 38 (parity) per cell; M4 skips the parity exam. Exams are one episode per pair, so cells are independent; Wilson 95% half-width ≈ ±0.15.

Arm	exam@0	exam@20	parity@0	parity@20
B0 (clean control)	—	0.718	—	0.579
M0 (full memory)	0.897	0.667	0.790	0.842
M2 (write-only)	0.692	0.718	0.526	0.605
M3 (+ corrective entry)	0.769	0.692	0.658	0.684
M4 (memory surgery)	0.744	0.667	—	

The experiment is pre-registered in the repository (`m1/PRE_REG_M1.md`, written before any M1 data was collected). Five arms: **B0**, 20 clean episodes with the same memory apparatus (equal-experience control); **M0**, 20 exposure episodes at ε=0.75 mix (six synthetic listings per page, same gated generator); **M2**, exposure with notes written but never injected back; **M3**, exposure plus an explicit corrective entry after every synthetic purchase; **M4**, starts from M0's end-of-exposure memory with all entries containing pre-registered price words deleted (18-word list in the preregistration); M4 skips the parity exam. Memory injection is recency-based (last three entries); lessons are distilled every five episodes by the same model. Exposure tasks are official goals [100:120); exams use lookalike pairs disjoint from every other experiment — discounted (f=0.9) pairs [250:290) and parity (f=1.0) pairs [290:330), 40 pairs each with fixed clone instances across arms and checkpoints. The exam pairs therefore never appear in any previously reported experiment (RA1 uses pairs [0:250)), and exposure tasks draw on official goals [100:120), a disjoint task space; the exam is thus leak-free by construction. 8 clone-generation failures are logged in the repository, leaving n=39 (exam) and n=38 (parity) as fixed exclusions across all arms. Checkpoints are clean episodes $\{0, 2, \ldots, 20\}$; exams are read-only. Seeds: exposure 50000+e, clean 60000+e, exam shuffles 41000+p, generation 99000+g.

A shared no-memory pre-exam gives $\sigma = 0.692$, and the B0 arm tracks it (0.718 at episode 2, p=0.80): the memory apparatus itself is neutral on the clean path. Pre-registered main tests: the internalization hypothesis on the discounted exam fails (0.667 vs. 0.718, p=0.62); the partial-decay branch holds (checkpoint-0 carryover McNemar 9/1, p=0.021; half-life 4 clean episodes); memory causality holds (write-only arm at chance, 0.692 vs. 0.897, $z = -2.24$, p=0.025). The parity persistence (0.842 vs. 0.579, p=0.011) and its memory mediation (0.790 vs. 0.526, p=0.016) are pre-registered secondary tests (single-seed findings; see below). The transfer matrix at checkpoint 0 is exploratory (n=20): same-category pairs show no elevation (0.718 vs. 0.744, p=0.80); a cross-category transfer was

not executable (all remaining pool pairs fall inside the five exposure categories — recorded as a feasibility limitation); the cross-domain praise-block test gives 1.00 vs. 0.70 (p=0.008) but the write-only (0.85) and surgery (1.00) arms are also elevated, so it is reported as exploratory with mixed evidence. The reward autopsy computes, for every exposure episode, the counterfactual official reward of the gold product: 38 synthetic purchases, 2 with reward at or above gold, mean gap −0.632.

Pre-registered re-randomizations (seeds 2028, 2029). Two replication runs shift the treatment-side randomness (exposure pools, episode dynamics, memory trajectories) while re-running the same five arms; in seed-2029 the exam instrument — instruction texts, clone candidates, and presentation order — is byte-identical to seed-2027, which restores the pre-exposure baseline (0.718 vs. seed-2027's 0.692; the fully shifted seed-2028 raises it to 0.821 because goal-pool shuffling changes 24 of 40 exam instructions). Pre-registered verdicts: the internalization failure replicates in both runs (p=0.53, p=0.62); the fast-channel carryover is direction-consistent (+3 and +5 pp) but not significant (p=1.0, p=0.75); the parity persistence and the write-only mediation contrast do not replicate ($p \geq 0.46$) and are reported as single-seed findings in the main text. All trajectories, pools, and verdict logs are released with the code.